Dynamic Cache Placement and Trajectory Design for UAV-Assisted Networks: A Two-Timescale Deep Reinforcement Learning Approach

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***Abstract*—Unmanned aerial vehicle (UAV)-assisted edge caching has been attracting significant attention, due to the potential of UAVs in delivering prompt content services to users. One of the key issues in UAV-assisted edge caching networks is the unpredictable user requests, which requires dynamic update of cache placement. Moreover, the timescale difference between cache placement strategy with long updating cycle and UAV trajectory planning strategy with short updating cycle makes them difficult to be optimized simultaneously. Against these backdrops, in this article, we consider a UAV-assisted edge caching network, where the UAVs act as aerial base stations (BSs), together with terrestrial BS to provide edge caching services for the mobile users. The user association, UAV cache placement and trajectory planning strategies are jointly optimized to maximize the throughput, while simultaneously satisfying the cache capacity and kinematics constraints of the UAVs. Specifically, we split the optimization problem into a cache placement subproblem in large timescale and a user association and trajectory planning subproblem in small timescale. A double deep Q-network (DDQN) method is adopted to solve the former subproblem, and an attention-based multi-agent deep deterministic policy gradient (ATMADDPG) algorithm is utilized to deal with the latter, in which the users and UAVs are regarded as heterogeneous agents to learn the accessing and flying policies for mutual benefits, respectively. Finally, a two-timescale deep reinforcement learning (2TDRL) algorithm is proposed to iteratively solve these two subproblems. Through comparisons with the other state-of-art schemes, simulation results verify the effectiveness of our proposed algorithm and demonstrate that a well-designed cache placement strategy can increase the throughput by 18.6%.**

***Index Terms*—Unmanned aerial vehicle (UAV)-assisted edge caching, user association, cache placement, UAV trajectory planning.**

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I. Introduction

RECENTLY, unmanned aerial vehicle (UAV) wireless networks [1] have been widely regarded as a promising way to complete the terrestrial communication networks and support the next generation wireless network, due to the higher link qualities, agile and resilient deployment ability, and flexible flying capability [2], [3]. In terms of applications, UAV wireless networks can be applied for a variety of scenarios, ranging from the ubiquitous coverage [4], relay [5] communication to data dissemination and collection [6]. For example, UAVs in [7] were deployed to learn the time-varying user activities to provide communication coverage, and UAVs in [8] were utilized to disseminate data to meet users' spatio-temporally distributed demands.

On the other hand, caching contents at the network edge has been envisioned as a promising solution to fulfill the users' dynamic requests, avoid duplicate data transmission and reduce the network latency [9], [10], [11]. Motivated by the benefits of UAV networks, there is increasing research attention in integrating the edge caching technology into the UAV networks [12], [13], [14], [15] in order to further relieve the heavy burden on the backhaul links [16], [17]. For example, in [12], a cache-enabled UAV communication network with underlaid user-to-user communications was considered, where the cache placement scheme was designed to maximize the cache hit probability. In [13], a UAV-enabled content-centric network was studied, where the file caching and retrieval costs were minimized by jointly optimizing file caching policy and communication scheduling. In [14], the cache-enabled UAV cellular network was investigated and the long-term caching placement, user scheduling and power allocation schemes were optimized to minimize the content delivery delay. In [15], a multi-UAV-enabled radio access network was considered and a hybrid caching and transmission strategy was proposed to maximize the overall spectral efficiency.

A. Related Works

In order to fully realize the potential of the UAV-assisted edge caching, the resource allocation and UAVs' trajectory planning need to be carefully designed, and extensive research attention has been devoted to the related topic [18], [19], [20], [21], [22], [23], [24], [25]. In terms of the methodologies, the research works can be categorized into the following two types: 1) the

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conventional optimization-based works; 2) the deep reinforcement learning (DRL)-based works.

The works based on the conventional optimization methods are firstly analyzed [18], [19], [20], [21]. For instance, in [18], the authors investigated the minimum user throughput maximization problem in a cache-enabled UAV-assisted network, where an alternating iterative algorithm was proposed to jointly optimize the UAV cache placement, trajectory planning and transmit power. In [19], the authors focused on the cache utility maximization problem in a UAV-device-to-device (D2D) network, where a matching and dynamic programming-based iterative algorithm was proposed to jointly optimize the cache placement and the UAV trajectory. In [20], the authors studied the throughput maximization problem in a Low Earth Orbit (LEO) satellite- and cache-enabled UAV communication system, where the block coordinate descent (BCD) and successive convex approximation (SCA) methods were applied to jointly optimize the UAV cache placement, trajectory planning and resource allocation under the constraints of limited cache capacity and flight time. In [21], the authors considered the minimum throughput maximization problem in the air-ground integrated network, where a cooperative dissemination algorithm was proposed to jointly optimize the user association, bandwidth allocation and UAV trajectory.

However, the conventional optimization methods are utilized to make decision for a specific environment and are limited by the high computational complexity, which is caused by repeatedly solving the problem at each time slot according to the frequently changed network topology. Furthermore, these algorithms often require the centralized controller and *a priori* environmental information (i.e., locations of users in all time slots), which are unlikely to be valid in practice.

To this end, the DRL-based algorithms [22], [23], [24], [25], [26] are considered to efficiently deal with the dynamic resource allocation and UAVs' trajectory planning scenarios with strong uncertainties. As a classic DRL algorithm, the deep Q-network (DQN) algorithm [22], [23] combines the advantages of the neural network and Q-learning, which utilizes a neural network to efficiently approximate the Q-value. For instance, in [22], the authors studied the energy-efficient content coverage problem, where a DQN-based algorithm was proposed to jointly optimize the trajectory design and recharging scheduling of multiple UAVs. In [23], the authors investigated the long-term throughput maximization problem in an aerial-terrestrial wireless D2D caching network, where a cooperative multi-agent DQN algorithm was proposed to jointly optimize the 3D trajectory planning of multiple UAVs. Further, in order to deal with the continuous actions, the actor-critic algorithm [24], [25], [26] has been proposed. For instance, in [24], the authors studied the energy consumption minimization problem in a dynamic multi-UAV-assisted MEC system, where the multi-agent deep deterministic policy gradient (MADDPG) algorithm was applied to jointly optimize the user association, resource allocation and trajectory planning. In [25], the authors investigated the cache hit ratio maximization problem in an air-ground caching network, where a cooperative multi-agent actor-critic-based DRL approach was designed to jointly optimize the cache placement and UAV deployment. In [26], the authors focused on the content acquisition delay minimization problem in a cellular wireless network, where a dual-clipped proximal policy optimization (PPO) algorithm was proposed to jointly optimize the user association, cache placement, UAV trajectory and transmission power.

B. Motivations and Contributions

In the UAV-assisted edge caching network, the limited caching capacity of UAVs makes the design of the cache placement strategy an important issue. However, the aforementioned works [22], [23], [24] only focused on the UAV trajectory planning with a fixed cache placement strategy. Moreover, the unpredictability in user requests is also one of the key issues that calls for dynamic update of cache placement. Although the authors in [18], [19], [20], [21], [25], [26] took both the cache placement and trajectory planning into consideration, they neglected the timescale difference between the cache placement strategy with long updating cycle and the UAV trajectory planning strategy with short updating cycle. The timescale difference makes them difficult to be optimized simultaneously. In addition, the unpredictable user requests, the changing content library, as well as the dynamic characteristic of the UAV networks produce great difficulties for the real-time resource scheduling and make the conventional optimization methods in [18], [19], [20], [21] increasingly less suitable for practical deployment. Moreover, in the DRL-based methods [22], [23], [24], [25], [26], the efficiency and scalability are hindered by the drastically increasing input dimension of the neural network.

Motivated by the above observations, in this article, we consider a UAV-assisted edge caching network, where the UAV base stations (BSs), together with the terrestrial BS, provide cache services for the mobile users. Special attention is payed to investigate the impact of dynamic user requests on the performance of UAV-assisted edge caching. To this end, we characterize the system throughput of UAV-assisted edge caching networks, reveling its dependence on user association, UAV cache placement and trajectory planning. To tackle the optimization problem, we then propose a two-timescale DRL (2TDRL) algorithm, which introduces the attention mechanism to reduce the exploration space and has shown to be particularly effective in solving problems with optimization variables of different timescale. The main contributions of this article are summarized as follows.

- We focus on the downlink transmission scenario in the UAV-assisted edge caching network, where the UAVs act as aerial BSs, together with the terrestrial BS, to provide prompt edge caching services for the mobile users. The throughput maximization problem is formulated by optimizing the UAV cache placement strategy with long updating cycle and the user association as well as trajectory planning strategies with short updating cycle, while simultaneously satisfying the cache capacity and kinematics constraints of the UAVs.
- Considering the timescale differences among the optimization variables, the optimization problem is further split into a cache placement subproblem in the large timescale and a user association and trajectory planning subproblem in the small timescale. A two-timescale 2TDRL algorithm is

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proposed to tackle these two subproblems hierarchically. Specifically, a double DQN (DDQN) method is applied to learn the cache placement directly based on the historical serving situations, which is effective to deal with the dynamically changing user requests and content library. Note that the double Q-learning is introduced to eliminate the overestimation. To deal with the latter subproblem, an attention-based MADDPG (A-MADDPG) algorithm is proposed, in which the attention mechanism is utilized to extract the key information as inputs to achieve more efficient performance in the complex environment of multiple agents, and the users and UAVs are regarded as heterogeneous agents to learn the accessing and flying policies for mutual benefits.

- Simulation results verify the effectiveness of our proposed algorithm compared with the state-of-the-art schemes, and demonstrate that a well-designed cache placement strategy can increase the throughput by 18.6%. Moreover, the cache storage capacity at each UAV needs to be judiciously selected to maximize the caching efficiency, and the optimal cache storage capacity is derived in the asymptotic scenarios with two ideal assumptions.

C. Organization

The rest of this article is organized as follows. Section II presents the system model of the considered UAV-assisted edge caching networks, including the user mobility, caching and communication models. Section III formulates the system throughput maximization problem. Section IV presents the details of our proposed 2TDRL algorithm. Numerical results and related discussions are provided in Section V. Finally, Section VI draws the conclusion. For clarity, the definitions of corresponding notations are summarized in Table I.

TABLE I
NOTATIONS OF PARAMETERS

Notation	Description
N, M	Number of UAVs and mobile users
T, K	Service cycle and number of slots in each cycle
H	The constant flight height of UAVs
S	Number of files in the file library
B	System bandwidth
$\mathbf{U}_n^k$, $\mathbf{M}_m^k$	3-D coordinate of UAV n and user m at time slot k
B_{BS}, B_n	Allocated band for the terrestrial BS and UAV n
v_m^k, θ_m^k	Speed and direction of user m at time slot k
c_n	Storage capacity of UAV n
$\mathcal{C}_n$	Cached file sets at UAV n
$p_{m,f}^k$	Probability of user m requesting file f_s at slot k
δ	Popularity factor
$\beta_{n,f}$	Cache placement indicator
$\alpha_{n,m,f}^k$	User association indicator
$H_{n,m}^k$, G_m^k	Channel coefficients from UAV n and BS to user m at time slot k, respectively
$h_{n,m}^k$, $L_{n,m}^k$	Small and large-scale fading
$\bar{h}_{n,m}^k$	Deterministic LoS channel component
$\hat{h}_{n,m}^k$	Random scattering component
$P_{n,m}^{k,LoS}$	The probability that the link between UAV n and user m at time slot k is LoS link
$p_{n,m}^k$, p_m^k	Transmit power of UAV n and terrestrial BS to user m at timeslot k, respectively
a_1, a_2	Pathloss exponent for the LoS and NLoS links in the U2D scenario, respectively
a_3	Pathloss exponent for links in B2U mode
$\gamma_{n,m}^{UAV}(k)$	SNR of user m served by UAV n at time slot k
$\gamma_m^{BS}(k)$	SNR of user m served by BS at time slot k
$R_m(k)$	Throughput of user m at time slot k
$R_{tot}(k)$	Throughput of all users at time slot k
$R_{uav}(k)$	Throughput of all UAVs at time slot k
$\mathcal{N}_t$	Exploration noise
p_b, p_c	Penalty for violating constraints (16f) and (16g), respectively
γ	Discount factor
$\mathcal{E}$	Size of replay buffer for the small timescale DRL
$\mathcal{G}$	Size of replay buffer for the large timescale DRL

II. System Model

In this article, we focus on the UAV-assisted edge caching network, in which the UAVs equipped with cache storage work as aerial BSs, together with a terrestrial BS, to provide content caching service for the mobile users. Note that these two types of entities complement each other and work together to provide users with prompt, all-round and seamless services. There are N UAVs and M mobile users in the system, represented by $\mathcal{N} = \{1, 2, \ldots, N\}$ and $\mathcal{M} = \{1, 2, \ldots, M\}$, respectively. The terrestrial BS is assumed to be located in the center of the geographical area. All the UAVs are in the coverage of the terrestrial BS and connected to it via wireless backhaul links. The service cycle of UAVs is set as T consisting of K equal time slots, as such, each time slot is $\tau = \frac{T}{K}$. τ is assumed to be small enough so that the velocity of UAVs remains constant within each time slot. Moreover, each UAV has the same height [1] and different two-dimensional (2D) coordinate, then the three-dimensional (3D) coordinate of UAV n, user m and terrestrial BS at time slot k can be expressed as $\mathbf{U}_n^k = (x_n^k, y_n^k, H)$, $\mathbf{U}_m^k = (x_m^k, y_m^k, 0)$ and $\mathbf{U}_0 = (x_0, y_0, 0)$, respectively. The system bandwidth is B, which can be further divided into orthogonal band resources B_{BS} and B_n for BS and UAV n to provide content transmission services to users, respectively.

[1]In this article, the distance between the UAVs and the users can be adjusted by the 2-D trajectory planning, and the barometer altitude determination method [31] is utilized in the UAVs' flight control system to keep their height fixed at H.

A. User Mobility Model

In this subsection, we present the user mobility model considered in this article. Specifically, the Gaussian Markov random model (GMSM) [32] is adopted to characterize the user mobility. Note that this model is widely used to describe the random movement of multiple users with similar speed and direction, mimicking the mobility patterns of users in search and rescue scenarios. As per the rules of GMSM, the speed and direction of user m at time slot k are, respectively, given by

$$v_m^k = \kappa_1 v_m^{k-1} + (1-\kappa_1)\,\bar{v} + \sqrt{1-\kappa_1^2}\Phi_m, \tag{1}$$

$$\theta_m^k = \kappa_2 \theta_m^{k-1} + (1-\kappa_2)\,\bar{\theta} + \sqrt{1-\kappa_2^2}\Psi_m, \tag{2}$$

where $0 \le \kappa_1 \le 1$ and $0 \le \kappa_2 \le 1$ are parameters that characterize the effect of the previous state on the current state, $\bar{v}$ and $\bar{\theta}$

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represent the average moving speed and direction of users in the system, respectively. In addition, Φ_m and Ψ_m follow the Gaussian distribution with different means and variances $(\bar{\zeta}_{v_n}, \sigma^2_{v_n})$ and $(\bar{\zeta}_{\theta_n}, \sigma^2_{\theta_n})$, respectively, representing the randomness of user movement. Based on (1) and (2), the horizontal coordinate of user m at time slot k is given by

$$x_m^k = x_m^{k-1} + v_m^{k-1}\tau\cos\left(\theta_m^{k-1}\right), \tag{3}$$

$$y_m^k = y_m^{k-1} + v_m^{k-1}\tau\sin\left(\theta_m^{k-1}\right). \tag{4}$$

B. Caching and Communication Model

In this subsection, we first provide the caching model of our work. Specifically, the file library consists of F equal sized files, represented by $\mathcal{F} = \{1, 2, \ldots, f, \ldots, F\}$. The storage capacity of UAV n is set as c_n, and the cached file sets of UAV n is represented by $\mathcal{C}_n$. We assume that the terrestrial BS contains the whole file library $\mathcal{F}$. We further assume that the probability of user m requesting for file f at time slot k is defined as $p^k_{m,f}$ and satisfies the Zipf distribution [38], given by,

$$p^k_{m,f} = \frac{f^{-\delta}}{\sum_{i=1}^{F} i^{-\delta}}, \tag{5}$$

where $0 < \delta < 1$ is the popularity factor, a bigger δ means that the distribution of request probability of files is more concentrated. Note that $\sum_{f=1}^{F} p^k_{m,f} = 1$. For the sake of simplicity, we assume that user m can only request for one file at a time.

We assume that the UAVs are able to proactively cache part of the files in $\mathcal{F}$ during off-peak hours. Specifically, in this article, we consider the cache-then-deliver scheme, namely the UAVs can perform the cache pre-placement before taking off. Note that the update of UAVs' cache placement can be based on the previously collected user requests. The cache storage of the UAVs are limited such that they cannot store all the files in $\mathcal{F}$. Therefore, when the files requested by the users are not cached in the UAVs, users will turn to the terrestrial BS for service. We define the cache placement indicator $\beta_{n,f}$, $\beta_{n,f} = 1$ means that UAV n cache the file f, and $\beta_{n,f} = 0$ means that UAV n doesn't cache the file f.

At the beginning of time slot k, each user decides whether to access the terrestrial BS or the UAVs to get the requested file. We introduce the binary variable $\alpha^k_{\bar{n},m,f}$ as a user association indicator, in which $\bar{n} \in \bar{\mathcal{N}} = \{0, 1, 2, \ldots, N\}$, and $\bar{n} = 0$ denotes the terrestrial BS. $\alpha^k_{\bar{n},m,f} = 1$ represents that user m accesses BS $\bar{n}$ for file f at time slot k. Note that each user can access to no more than one BS, while each BS is able to serve multiple users.

In the following, we detail the channel model considered for the UAV to user (U2D) scenario and the terrestrial BS to user (B2D) scenario, respectively.

We first focus on the U2D channel model. The channel coefficient between UAV $n \in \mathcal{N}$ and user m at time slot k is given by

$$H^k_{n,m} = h^k_{n,m} L^k_{n,m}, \tag{6}$$

where $h^k_{n,m}$ represents the small-scale fading. In this article, we assume $h^k_{n,m}$ follows the Rician distribution [33], which is given by

$$h^k_{n,m} = \left(\sqrt{\frac{Q^k_{n,m}}{Q^k_{n,m}+1}}\bar{h}^k_{n,m} + \sqrt{\frac{1}{Q^k_{n,m}+1}}\hat{h}^k_{n,m}\right), \tag{7}$$

where $\bar{h}^k_{n,m}$ is the deterministic LoS channel component and $|\bar{h}^k_{n,m}| = 1$. $\hat{h}^k_{n,m}$ is the random scattering component, which is a circularly symmetric complex Gaussian (CSCG) random variable with zero mean and unit variance [34]. $Q^k_{m,n}$ denotes the Rician factor.

In addition, $L^k_{n,m}$ represents the large-scale fading, which is expressed as [35]

$$L^k_{n,m} = \sqrt{P^{k,LoS}_{n,m}\beta_0\left(d^k_{n,m}\right)^{-a_1} + \left(1 - P^{k,LoS}_{n,m}\right)\kappa\beta_0\left(d^k_{n,m}\right)^{-a_2}}, \tag{8}$$

where β_0 is the power gain at reference distance $d_0 = 1$ m, a_1 and a_2 are the path loss coefficients for LoS and NLoS links, respectively, $0 < \kappa < 1$ is the regularized factor due to NLoS links, and $P^{k,LoS}_{n,m}$ is the probability that the link between UAV n and user m at time slot k is a LoS link. In (8), $P^{k,LoS}_{n,m}$ is expressed as [36]

$$P^{k,LoS}_{n,m} = \frac{1}{1 + b_1\exp\left(-b_2\left[\arcsin\left(\frac{H}{d^k_{n,m}}\right) - b_1\right]\right)}, \tag{9}$$

where $d^k_{n,m} = \sqrt{\left(x^k_n - x^k_m\right)^2 + \left(y^k_n - y^k_m\right)^2 + H^2}$ is the distance between UAV n and user k at time slot k, b_1 and b_2 are constants related to the propagation environment.

According to (6)–(9), the signal-to-noise ratio (SNR) of user m served by UAV n at time slot k is given by

$$\gamma^{UAV}_{n,m}(k) = \frac{p^k_{n,m}\left|H^k_{n,m}\right|^2}{\left(\sigma^k_{n,m}\right)^2}, \tag{10}$$

where $p^k_{n,m}$ is the downlink transmit power of UAV n to user m at time slot k, and $\sigma^k_{n,m} = N_0 B^k_{n,m}$ with N_0 denoting the noise power spectrum density and $B^k_{n,m}$ denoting the band resource allocated to user m served by UAV n at time slot k. For the sake of simplicity, we assume that the bandwidth B_n is equally allocated to its serving users. As such, we have $B^k_{n,m} = \frac{B_n}{\sum_{m=1}^{M}\sum_{f=1}^{F}\alpha^k_{n,m,f}}$.

Therefore, the transmission rate of user m served by UAV n at time slot k is given by

$$R^{UAV}_{n,m}(k) = \sum_{f=1}^{F}\alpha^k_{n,m,f}\beta_{n,f}B^k_{n,m}\log_2\left(1 + \gamma^{UAV}_{n,m}(k)\right). \tag{11}$$

Then, we focus on the B2D channel model. We consider that the channels between the terrestrial BS and user m at time slot k are subject to the Rayleigh fading [37], given by

$$G^k_m = \sqrt{|g^k_m|^2\left(d^k_m\right)^{-a_3}}, \tag{12}$$

where g^k_m are independent and identically distributed complex Gaussian random variables with zero mean and unit variance.

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$d_m^k = \sqrt{(x_0 - x_m^k)^2 + (y_0 - y_m^k)^2}$ is the distance between the terrestrial BS and user m at time slot k, and a_3 is the path loss coefficient.

Based on (12), the SNR of user m served by terrestrial BS at time slot k is given by

$$\gamma_m^{BS}(k) = \frac{p_{0,m}^k \left|G_m^k\right|^2}{\left(\sigma_m^k\right)^2}, \tag{13}$$

where $p_{0,m}^k$ is the downlink transmit power of terrestrial BS to user m at time slot k, $\sigma_m^k = B_m^k N_0$ with B_m^k denoting the bandwidth resource allocated to user m served by the terrestrial BS at time slot k. As such, we have $B_m^k = \frac{B_{BS}}{\sum_{m=1}^{M}\sum_{f=1}^{F}\alpha_{0,m,f}^k}$. Then, the transmission rate of the terrestrial BS can be expressed as

$$R_m^{BS}(k) = \sum_{f=1}^{F} \alpha_{0,m,f}^k B_m^k \log_2\left(1 + \gamma_m^{BS}(k)\right). \tag{14}$$

III. Problem Formulation

Based on the analysis in Section II, the throughput of user m at time slot k is given by

$$R_m(k) = R_m^{BS}(k) + \sum\nolimits_{n=1}^{N} R_{n,m}^{UAV}(k). \tag{15}$$

In this article, our goal is to maximize the system throughput by jointly optimizing the user association strategy $\boldsymbol{\alpha} = \{\alpha_{\bar{n},m,f}^k | \forall \bar{n} \in \bar{\mathcal{N}}, \forall m \in \mathcal{M}, \forall f \in \mathcal{F}\}$, the UAV cache placement strategy $\boldsymbol{\beta} = \{\beta_{n,f} | \forall n \in \mathcal{N}, \forall f \in \mathcal{F}\}$, and the trajectory planning strategy $\mathbf{U} = \{\mathbf{U}_n^k | \forall n \in \mathcal{N}\}$, while simultaneously satisfying the cache capacity and kinematic constraints of the UAVs. The optimization problem is given by

$$\begin{aligned}
\max_{\boldsymbol{\alpha},\boldsymbol{\beta},\mathbf{U}} \quad & \sum_{k=1}^{K} R_{tot}(k) \\
s.t. \quad & \alpha_{\bar{n},m,f}^k \in \{0,1\}, (a) \\
& \sum_{n=0}^{N}\sum_{f=1}^{F} \alpha_{\bar{n},m,f}^k \le 1, (b) \\
& \beta_{n,f} \in \{0,1\}, (c) \\
& \sum_{f=1}^{F} \beta_{n,f} \le c_n, (d) \\
& \tau V_{\min} \le \left\|\mathbf{U}_n^{k+1} - \mathbf{U}_n^k\right\| \le \tau V_{\max}, (e) \\
& \mathbf{U}_L \le \mathbf{U}_n^k \le \mathbf{U}_U, (f) \\
& \left\|\mathbf{U}_n^k - \mathbf{U}_i^k\right\| \ge d_{col}, (g)
\end{aligned} \tag{16}$$

where $R_{tot}(k) = \sum_{m=1}^{M} R_m(k)$. Constraints (16a) and (16b) are the user association constraints, indicating that user m can access to no more than one BS at each time slot. Constraint (16c) regulates that the cache placement variable is binary, constraint (16d) means that the number of files cached in UAV n should not exceed its storage. Constraint (16e) gives the range of UAVs' flying speed, where v_{min} and v_{max} are the minimum and maximum speed, respectively. Constraint (16f) regulates the feasible flying range of UAVs. Constraint (16g) is the collision avoidance constraint between any two UAVs, where d_{col} is the minimum tolerable distance.

Note that the optimization problem (16) is a mixed integer non-linear programming (MINLP) problem due to the discrete and continuous variables as well as the non-convex objective function and constraints, which is non-deterministic polynomial-time hard (NP-hard) [39] and cannot be solved directly. Therefore, in the following subsection, we choose to decompose the original problem (16) to facilitate the settlement.

A. Users' and UAVs' Perspectives

To this end, we can split the considered optimization problem into two subproblems from the users' and the UAVs' perspectives, respectively. From the perspective of users, the goal is to obtain the cache service by judiciously selecting the serving BS, so as to maximize the system throughput. The corresponding optimization problem of users can be formulated as follows [2]

$$\begin{aligned}
\max_{\boldsymbol{\alpha}} \quad & \sum_{k=1}^{K} R_{tot}(k) \\
s.t. \quad & (16a)(16b).
\end{aligned} \tag{17}$$

From the perspective of UAVs, the goal is to provide users with high-quality cached file transmission services through cache placement and trajectory planning strategies, so as to maximize the system throughput. The corresponding optimization problem of UAVs can be formulated as follows

$$\begin{aligned}
\max_{\boldsymbol{\beta},\mathbf{U}} \quad & \sum_{k=1}^{K} R_{tot}(k) \\
s.t. \quad & (16c) - (16g).
\end{aligned} \tag{18}$$

Based on (15) we can see that the variables $\boldsymbol{\beta}$ and $\mathbf{U}$ are irrelevant to the throughput contributed by the terrestrial BS, as such, the objective function in (18) can be further simplified and the optimization problem can be rewritten as

$$\begin{aligned}
\max_{\boldsymbol{\beta},\mathbf{U}} \quad & \sum_{k=1}^{K} R_{uav}(k) \\
s.t. \quad & (16c) - (16g),
\end{aligned} \tag{19}$$

where $R_{uav}(k) = \sum_{n=1}^{N}\sum_{m=1}^{M} R_{n,m}^{UAV}(k)$.

Based on the optimization problems (17) and (19), we find that there exists interaction between the users and the UAVs. In particular, the UAVs' cache placement and trajectory planning strategies in (19) can affect the user association strategy in (17). Similarly, different user associations will lead to different UAV cache placement and trajectory planning strategies.

B. Timescale-Based Hierarchical Decomposition of the Formulated Optimization Problem in (19)

In this subsection, we first analyze the timescale difference for various optimization variables in (19). Then, the timescale-based hierarchical decomposition of the optimization problem in (19) is further performed based on this difference.

[2]Note that (17) is a linear integer programming problem of $\alpha_{\bar{n},m,f}^k$, which is feasible. While due to the exponentially increasing computational complexity $\mathcal{O}((N+1)^M)$, it is hard to be directly solved by the conventional optimization methods. In the following, we apply the DRL method to achieve the comparable performance while inducing a lower complexity.

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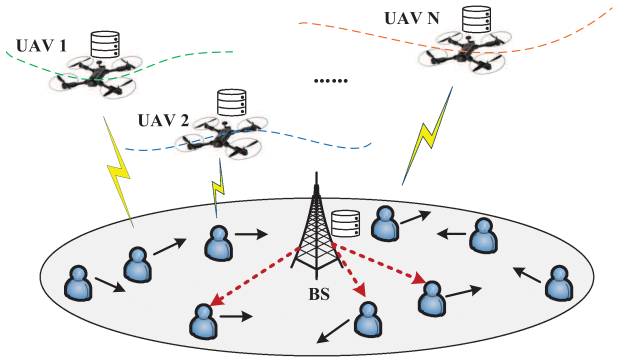


Fig. 1. Illustration of the UAV-assisted edge caching network with mobile users.

Note that the update cycle of the cache placement strategy is different from the user association strategy and the trajectory planning strategy. Specifically, the UAVs cache relevant files before the flight, and the cache placement strategy remains unchanged throughout the whole flight. However, the user association and UAV trajectory planning strategies change every time slot during the flight cycle. Therefore, the optimization problem in (19) needs to be further decomposed into two timescales. That is, the outer large timescale with long update cycle and the inner small timescale with short update cycle, which are respectively given by

$$\max_{\boldsymbol{\beta}} \quad \sum_{k=1}^{K} R_{uav}(k) \qquad s.t. \quad (16c)(16d). \tag{20}$$

$$\max_{\mathbf{U}} \quad \sum_{k=1}^{K} R_{uav}(k) \qquad s.t. \quad (16e)-(16g). \tag{21}$$

We call the long and short update cycles as the time frame and the time slot, respectively. Note that one time frame is equivalent to a flight cycle, which consists of K time slots. The cache placement, user association and UAVs' trajectory planning are optimized sequentially. Specifically, at the beginning of each time frame, each UAV decides the cache placement strategy based on (20) and caches corresponding files into its storage. At the beginning of each time slot, each user selects a UAV to access based on (17), and each UAV moves to the next position based on (21). Note that during the K time slots, user association and UAV trajectory planning strategies are optimized with the given cache placement strategy obtained in the last time frame. The illustration of the timescale-based decomposition is shown in Fig. 2.

Due to the above observations, together with the facts that the considered scenarios are dynamic and the variables are different in timescale, traditional optimization-based approaches are not suitable anymore, since they generally require a central controller and *a priori* environmental information (i.e., the locations of users in all time slots), and induce high computational complexity. To address this issue, in the next section, we leverage the framework of DRL and multi-agent DRL (MADRL) [40], [41] methods, which can not only tackle with the timescale difference of optimization variables, but also simultaneously enable the users and the UAVs to learn the user association policies and the UAVs' cache placement and trajectory planning policies, such that the problems can be solved efficiently with significantly less computational complexity.

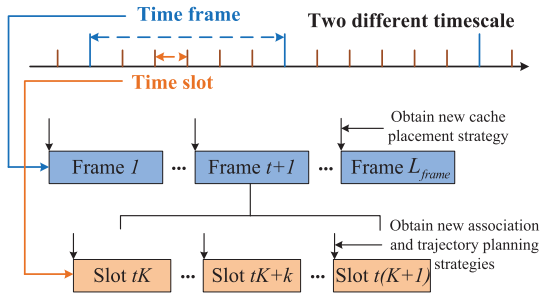


Fig. 2. Illustration of the timescale-based decomposition.

IV. Two-Timescale Deep Reinforcement Learning for UAV-Assisted Edge Caching Networks

By utilizing DRL, in this section, we propose a two-timescale 2TDRL algorithm to determine the cache placement, user association, and trajectory planning strategies. In particular, as shown in Figs. 2 and 3, to deal with the small timescale user association and UAV trajectory planning optimization problems in (17) and (21) that updated per time slot, we propose an attention-based MADDPG algorithm, in which the attention mechanism is introduced to better extract the key information as inputs to achieve more efficient performance in the complex environment of multiple agents, and the users and UAVs are regarded as heterogeneous agents to learn the accessing and flying policies for mutual benefits. To deal with the large timescale cache placement optimization problem in (20) that updated per time slot, we propose a DDQN algorithm to dynamically adjust UAVs' cached files based on the historical users' requests, and the double Q-learning method is introduced to eliminate the overestimation. Based on the above, the two problems with different timescales are tackled hierarchically in the proposed 2TDRL algorithm.

A. User Association and UAV Trajectory Planning Problem in Small Timescale

For a given cache placement $\boldsymbol{\beta}$ that obtained in the last time frame, the users and the UAVs act as two different types of agents who want to learn which BS to access and which location to move to all by themselves, respectively, in order to solve (17) and (21). In the following, we will first describe the elements of the MADRL problem at the UAV and user agents, and then give a basic introduction of the MADDPG algorithm. Finally, we adopt the attention mechanism to further enhance the efficiency of MADDPG.

1) Elements of the MADRL Problem: As shown in Fig. 3, the elements of the MADRL problem at each agent are described as follows.

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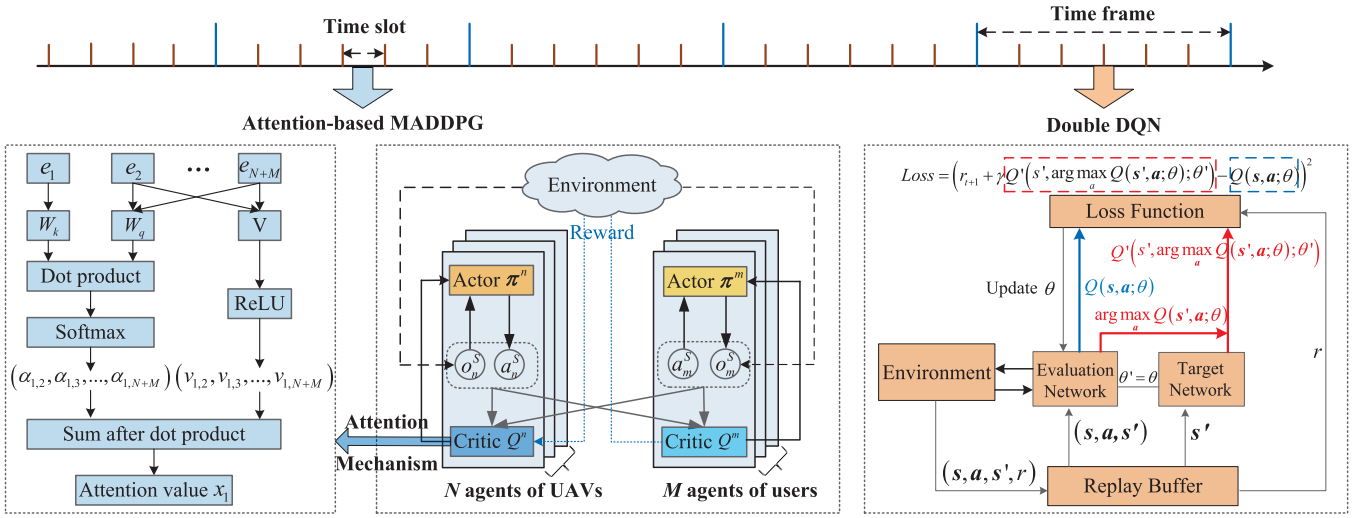


Fig. 3. Overall algorithm flowchart of the proposed two timescale 2TDRL scheme.

- Observation Space: *For UAV agent:* The observation of UAV n at time slot k can be expressed as follows

$$\boldsymbol{o}_n^S(k) \triangleq \left\{\boldsymbol{v}_n^k, \left\{\mathbf{U}_n^k\right\}_{n\in\mathcal{N}}, \left\{\mathbf{M}_m^k\right\}_{m\in\mathcal{M}}\right\}, \tag{22}$$

which includes the current velocity and location of UAV n, locations of users and locations of all the other UAVs. In particular, we use $\boldsymbol{v}_n^k$ to 6represent the flying velocity of UAV n at time slot k and we have $\boldsymbol{v}_n^k = [v_{n,x}^k, v_{n,y}^k]$. We assume that the UAVs can exchange information with each other to ensure that any one of them know the location information of all UAVs and users. As such, there are $2(1+N+M)$ elements contained in the observation $\boldsymbol{o}_n^S(k)$ of UAV n at time slot k.

For user agent: We assume that there exists no information exchange among users to ensure the user privacy. As such, the observation of user m at time slot k only includes its local information and can be expressed as

$$\boldsymbol{o}_m^S(k) \triangleq \left\{\left\{\sum\nolimits_{f\in\mathcal{F}} \alpha_{\bar{n},m,f}^{k-1}\right\}_{\bar{n}\in\bar{\mathcal{N}}}, \left\{\Delta p_{\bar{n},m}^k\right\}_{\bar{n}\in\bar{\mathcal{N}}}, s_m^k\right\}, \tag{23}$$

which includes user m's connected BS $\bar{n}$ at the previous time slot $k-1$, the received power gap between the two consecutive time slots, and its requested file at the current time slot k, $s_m^k \in \mathcal{F}$. There are $2N+1$ elements contained in the observation $\boldsymbol{o}_m^S(k)$ of user m at time slot k.

- Action Space: *For UAV agent:* The action of UAV n at time slot k can be expressed as

$$\mathbf{a}_n^S(k) \triangleq \left\{\boldsymbol{v}_n^k\right\}, \tag{24}$$

where $\|\mathbf{a}_n^S(k)\| = \sqrt{(v_{n,x}^k)^2 + (v_{n,y}^k)^2} \in [v_{min}, v_{max}]$. The coordinate of UAV n is $\mathbf{U}_n(k) = (x_n^k, y_n^k, H)$ with $x_n^k = x_n^{k-1} + v_{n,x}^k \tau$ and $y_n^k = y_n^{k-1} + v_{n,y}^k \tau$.

For user agent: The action of user m at time slot k can be expressed as

$$\mathbf{a}_m^S(k) \triangleq \left\{\alpha_{\bar{n},m,f}^{k-1}\right\}_{\bar{n}\in\bar{\mathcal{N}}, f\in\mathcal{F}}. \tag{25}$$

- Reward Function: *For UAV agent:* The reward function of UAV n at time slot k is given by

$$r_n^S(k) = R_{uav}(k) - \varphi_b(k) p_b - \varphi_c(k) p_c. \tag{26}$$

It is worth noting that the constraints (16f) and (16g) have to be taken into consideration when we design the reward function. We introduce the penalty indicator $\varphi_b(k)$ and $\varphi_c(k)$. When the flying range constraint (16f) is unsatisfied, $\varphi_b(k) = 1$, and penalty p_b is used to punish the UAV. Similarly, when the collision avoidance constraint (16g) is unsatisfied, $\varphi_c(k) = 1$, and penalty p_c is used to punish the UAV.

For user agent: According to (17), the reward function of user m at time slot k is defined as $r_m^S(k) = R_{tot}(k)$.

2) MADDPG Approach: In this article, we exploit the framework of MADDPG algorithm [27], [28] to deal with the optimization problems in (17) and (21). Specifically, in the training phase of MADDPG, agents can utilize actions and observations of other agents to facilitate the training. In the execution phase, agents have to make policy decisions based on their local observations. The actor and the critic networks for each agent i can be expressed as $\pi_i^S(\mathbf{o}_i^S; \theta_i^\pi)$ and $Q_i^S(\mathbf{o}, \mathbf{a}; \theta_i^Q)$, respectively, in which $\mathbf{o}(k) = \{\boldsymbol{o}_1^S(k), \boldsymbol{o}_2^S(k), \ldots, \boldsymbol{o}_I^S(k)\}$ and $\mathbf{a}(k) = \{\mathbf{a}_1^S(k), \mathbf{a}_2^S(k), \ldots, \mathbf{a}_I^S(k)\}$ are the joint observation and action sets of all agents. θ_i^π and θ_i^Q are parameters for actor and critic networks, respectively.

To enhance the stability, we further adopt the target network and experience replay technologies. Specifically, for each agent i, we introduce two extra target networks $\pi_i^{S'}(\mathbf{o}_i^S; \theta_i^{\pi'})$ and $Q_i^{S'}(\mathbf{o}, \mathbf{a}; \theta_i^{Q'})$. The corresponding experience tuples $(\mathbf{o}(k), \mathbf{a}(k), \mathbf{r}(k), \mathbf{o}(k+1))$ are stored in the experience replay buffer $\mathcal{E}$. When $\mathcal{E}$ is full, the original tuples in the replay buffer will be replaced by the new generated ones.

The critic network is updated based on the minimization of the mean square error (MSE) loss

$$L\left(\theta_i^Q\right) = \frac{1}{N_B} \sum_k \left[y_i^S(k) - Q_i^S\left(\mathbf{o}(k), \mathbf{a}(k); \theta_i^Q\right)\right]^2, \tag{27}$$

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with the target value $y_i^S(k)$ given by

$$y_i^S(k) = r_i^S(k) + \gamma Q_i^{S'}\left(\mathbf{o}'(k+1), \mathbf{a}'(k); \theta_i^{Q'}\right), \quad (28)$$

where γ is the discount factor, N_B is the batch size, $\mathbf{a}'(k) = \{\mathbf{a}_1^{S'}(k), \ldots, \mathbf{a}_I^{S'}(k)\}$, and $\mathbf{a}_i^{S'}(k) = \pi_i^{S'}(\mathbf{o}(k); \theta_i^{\pi'})$.

The actor network is updated based on the sampled policy gradient, which is given by

$$\nabla_{\theta_i^\pi} J\left(\pi_i^S\right) \approx \frac{1}{N_B} \sum_k \nabla_{\theta_i^\pi} \pi_i^S\left(\boldsymbol{o}_i(k)\right) \nabla_{a_i} Q_i^S\left(\mathbf{o}(k), \mathbf{a}(k); \theta_i^Q\right). \quad (29)$$

Finally, for the given θ_i^π and θ_i^Q of the evaluation networks, we adopt the soft updating strategy and the parameters of target networks are updated as

$$\theta_i^{\pi'} = \xi^{act}\theta_i^\pi + \left(1 - \xi^{act}\right)\theta_i^{\pi'}, \quad (30)$$

$$\theta_i^{Q'} = \xi^{cri}\theta_i^Q + \left(1 - \xi^{cri}\right)\theta_i^{Q'}, \quad (31)$$

where $\xi_n^{act} \ll 1$ and $\xi_n^{cri} \ll 1$.

3) Attention-Based MADDPG Approach: As shown in the left of Fig. 3, the attention mechanism [29], [30] is introduced to reduce the exploration space and enhance the learning efficiency. Note that in conventional MADDPG, the inputs of the critic network for each agent i are the observations $\mathbf{o}$ and actions $\mathbf{a}$ of all the agents. When the number of agents increases, the dimensionality of the inputs will increase drastically, making it more and more difficult for the neural network to learn and scale up. By using the attention mechanism, the most important information of the observation-action pairs can be extracted as the inputs of each agent's critic network, which can not only solve the scalability problem, but also help the agents to selectively filter the important environmental information and ignore the unimportant. Specifically, in the attention-based MADDPG approach, the Q-function of agent i can be rewritten as

$$Q_i^S(\mathbf{o}, \mathbf{a}) = f_i\left(g_i\left(o_i^S, a_i^S\right), x_i\right), \quad (32)$$

where $f_i(\cdot)$ is a four-layer multi-layer perception (MLP) and $g_i(\cdot)$ is a one-layer MLP for the agent i. x_i is the attention value, which is given by

$$x_i = \sum\nolimits_{j \neq i} \mu_{ij} v_{ij} = \sum\nolimits_{j \neq i} \mu_{ij} h\left(V g_j\left(o_j^S, a_j^S\right)\right), \quad (33)$$

where μ_{ij} represents the attention weight of the agent i to the agent j. Here we denote $e_i = g_i(o_i^S, a_i^S)$, and μ_{ij} can be calculated by comparing the similarity between e_i and e_j, which is given by

$$\mu_{ij} = \mathrm{softmax}\left(e_j^T W_k^T W_q e_i\right), \quad (34)$$

where the parameter matrices W_q and W_k are used to map e_i and e_j into query and key, respectively, and the $\mathrm{softmax}(\cdot)$ operation is used to get the normalized attention weight.

In addition, v_{ij} in (33) represents the embedding operation on agent j's observation-action pair and in particular, V is the transformed matrix to perform the linear transformation, $h(\cdot)$ is a rectified linear unit (ReLU) function. In this article, we adopt multi-head attention mechanism, where the attention value x_i for different agent i is calculated using different parameter matrices (W_q, W_k, V).

To summarize, by inputting agent i's own observation-action pair and the attention value x_i into the critic network, agent i is able to pay more attention to the similar agents, which results in a higher utilization of information. Moreover, by combining the observation-action pairs of all agents into a single dimensional vector, the scalability problem can also be solved.

B. UAV Cache Placement Problem in Large Timescale

For the optimization problem in (20), the cache placement strategy of UAVs should be elaborately designed to maximize the throughput. Note that the cache placement is updated per time frame. When each flight cycle is completed, the UAV ground control station (UGCS) updates the cache placement of all the UAVs based on the feedback of the last service cycle. For the sake of simplicity, in this article, the terrestrial BS also works as the UGCS. The elements of the DRL problem at UGCS is described as follows.

1) Observation Space: The observation at time frame t is

$$\mathbf{o}_t^F = \left\{\left\{\mathbf{q}_n^t\right\}_{n \in \mathcal{N}}\right\}, \quad (35)$$

where $\mathbf{q}_n^t = [q_{n,1}^t, q_{n,2}^t, \ldots, q_{n,F}^t]$, and $q_{n,f}^t$ represents the file popularity of file f observed by UAV n at time frame t, which is defined as the average times of file f requested by users during the previous K time slots. $q_{n,f}^t$ is given by

$$q_{n,f}^t = \frac{\sum_{i=(t-1)K+1}^{tK} \sum_{n=1}^{N} \sum_{m=1}^{M} \alpha_{n,m,f}^i}{K}. \quad (36)$$

2) Action Space: The action at time frame t is

$$\mathbf{a}_t^F = \left\{\left\{\boldsymbol{\beta}_n\right\}_{n \in \mathcal{N}}\right\}, \quad (37)$$

where $\boldsymbol{\beta}_n = [\beta_{n,1}, \beta_{n,2}, \ldots, \beta_{n,F}]$.

3) Reward Function: At the beginning of each time frame, all UAVs take the action of cache placement based on the observation $\mathbf{o}^F(t)$. Once this time frame ends, the corresponding throughputs during K slots of the whole time frame will be reported back. Then the reward can be computed as the average of these feedbacks, which is given by

$$r_t^F = \frac{\sum_{i=(t-1)K+1}^{tK} R_{uav}(i)}{K}. \quad (38)$$

In this subsection, a deep Q-learning (DQN) algorithm is utilized to solve the optimization problem (20). In this algorithm, the UGCS agent maintains a deep Q-network $Q^F(\mathbf{o}_t^F, \mathbf{a}_t^F; \theta^F)$ that takes the current state $\mathbf{o}_t^F$ as an input and output Q-values of all related actions. θ^F is the network weight. The ϵ-greedy policy is adopted to balance the exploration and exploitation. To enhance the stability of this algorithm, the target network and experience replay technologies are also applied. After each time frame, the tuple $\{\mathbf{o}_t^F, \mathbf{a}_t^F, r_t^F, \mathbf{o}_{t+1}^F\}$ is stored in the replay buffer $\mathcal{G}$.

Note that in the traditional DQN algorithm, $\max_a Q(s', a; \theta)$ is used to update the value of action, which may lead to a maximization bias. To solve the overestimation problem in DQN, double Q-learning is applied in DQN to get the DDQN network, where the selection and the evaluation of actions in the target Q value are separated. Specifically, in DDQN, the evaluation network is used to determine the action, and the target network

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Algorithm 1: 2TDRL Algorithm.

Input:
learning rate, episode length L_{ep}, time frame length L_{frame}, discount factor γ, buffer size, mini-batch size, penalties;
Output:
Well-trained neural network parameters;
1: Initialize the replay buffer $\mathcal{G}$ and $\mathcal{E}$;
2: Initialize the neural and target neural networks of each agent with the corresponding weights;
3: **for** episode=1: L_{ep} **do**
4: Reset the environment;
5: **for** time frame t=1: L_{frame} **do**
6: Observe the environment to obtain $\mathbf{o}_t^F$;
7: The UGCS agent randomly chooses the cache place-ment action for all the UAVs with probability ϵ or otherwise takes the action with maximum Q value;
8: Each UAV n gets informed of its cache placement decision for the coming flight;
9: **for** time slot k=1:K **do**
10: Each UAV agent n and each user agent m obtain the observations and choose corresponding moving and accessing actions, respectively;
11: All agents take actions and get rewards;
12: **if** UAV agent n violate constraints (16 f) or (16 g)
13: Punish the UAV and cancel its movement;
14: **end if**
15: All agents get their rewards $r_n^S(k)$ and $r_m^S(k)$, and observations of next slot $\mathbf{o}_n^S(k+1)$ and $\mathbf{o}_m^S(k+1)$;
16: Store tuples $(\mathbf{o}(k), \mathbf{a}(k), \mathbf{r}(k), \mathbf{o}(k+1))$ into $\mathcal{E}$;
17: **if** $\mathcal{E}$ is full **then**
18: Randomly sample the experience tuples from $\mathcal{E}$ to form the mini-batch;
19: All agents set their target value according to (28), and update the critic and actor networks based on (27) and (29), respectively;
20: **end if**
21: Soft update the weights of the target networks according to (30) and (31);
22: **end for**
23: The UGCS agent computes the reward r_t^F based on the average of rewards that the UAVs obtained in the previous K time slots, and gets the observation of next frame $\mathbf{o}_{t+1}^F$;
24: Store the tuple $\{\mathbf{o}_t^F, \mathbf{a}_t^F, r_t^F, \mathbf{o}_{t+1}^F\}$ into $\mathcal{G}$;
25: **if** $\mathcal{G}$ is full **then**
26: Randomly sample experience tuples from $\mathcal{E}$ to form the mini-batch;
27: Update the network by minimizing (40);
28: **end if**
29: Soft update the weight of the target Q-network;
30: **end for**
31: **end for**

is used to determine the value of the action. The target value is given by

$$y_t = r_t^F + \gamma Q'\left(\mathbf{o}_{t+1}^F, \arg\max_a Q\left(\mathbf{o}_{t+1}^F, \mathbf{a}_t^F; \theta^F\right); \bar{\theta}^F\right), \quad (39)$$

where $\bar{\theta}^F$ is the weight of the target network. A mini-batch of data is sampled from $\mathcal{G}$ to update θ^F by minimizing the following loss function

$$L\left(\theta^F\right) = \left(y_t - Q\left(\mathbf{o}_t^F, \mathbf{a}_t^F; \theta^F\right)\right)^2. \quad (40)$$

Note that the cache placement policy of each UAV is randomly initialized, and gradually enhanced to the optimal policy with the update of DDQN.

C. Two-Timescale 2TDRL Algorithm

Based on the large timescale user association and UAV trajectory planning problem solved in Section IV.A and the small timescale cache placement problem solved in Section IV.B, we finally propose a novel 2TDRL algorithm, in which the two problems are hierarchically optimized. The algorithm procedure is summarized in Algorithm 1.

Specifically, at the beginning of each time frame t, the UGCS agent n adopts the ϵ-greedy policy to randomly caches content files in UAVs with probability ϵ or otherwise takes the cache placement action with maximum Q value. The cache placement strategy remains unchanged during the next K time slots. Therefore, with the given $\boldsymbol{\beta}(t)$, at the beginning of each time slot k, each user agent m and each UAV agent n get their observations of the environment, and choose corresponding accessing action $\mathbf{a}_m^S(k)$ and moving action $\mathbf{a}_n^S(k)$, respectively. Note that the exploration noise $\mathcal{N}_t$ is added to the action, and the exploration-exploitation balance is achieved by tuning it, which is a Gaussian random variable with zero mean and unit variance. After all agents have performed the actions, the rewards $r_m^S(k)$ and $r_n^S(k)$ can be obtained and the system moves to the next state. Note also that the action of UAVs should not violate constraints (16f) or (16g), otherwise, the UAVs should be punished and the movements should be cancelled. Then the tuple $\{\boldsymbol{o}_i^S(k), \mathbf{a}_i^S(k), r_i^S(k), \boldsymbol{o}_i^S(k+1)\}, i \in \{\mathcal{M} \cup \mathcal{N}\}$ is stored in the replay buffer $\mathcal{E}$. When $\mathcal{E}$ is full, we sample a mini-batch from that and update parameters of critic and actor networks based on (27) and (29), respectively. In addition, using the target networks mentioned in Section IV-A, the weights of the target networks are updated based on (30) and (31). When the K slots end, the UGCS agent computes the reward of the time frame t based on the average of rewards that obtained in previous K time slots. Similarly, when $\mathcal{G}$ is full, we sample a mini-batch from that and update parameters of DDQN by minimizing (40).

After the training process is completed, the obtained algorithm can enable UAVs and mobile users to make intelligent user association, cache placement and trajectory planning decisions that is adaptive to dynamic user requests in a distributed way during the execution process.

D. Complexity Analysis

As shown in the overall flowchart in Fig. 3, the proposed 2TDRL algorithm consists of the outer DDQN algorithm in the large timescale and the inner A-MADDPG algorithm in the small timescale. Firstly, we analyze the computational complexity of the DDQN algorithm. As referred in [44], we assume that the neural network contains L fully connected layers, and u_l is the number of neurons in the l-th layer.

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For a single sample, the computational complexity of each time step can be denoted as $\mathcal{O}(\sum_{l=0}^{L-1} u_l u_{l+1})$. Thus, for L_{ep} episodes and L_{frame} time frames, the computational complexity of the outer DDQN algorithm is $\mathcal{O}(L_{ep}L_{frame}\mathcal{X})$, in which $\mathcal{X} = \sum_{l=0}^{L-1} u_l u_{l+1}$. Similarly, for the A-MADDPG algorithm, as referred in [45], we assume that the actor network and the critic network contain A and C fully connected layers, respectively, and the number of neurons in the a-th layer of the actor network and in the c-th layer of the critic network are u_a^{actor} and u_c^{critic}, respectively. For a single sample, the computational complexity of each time step can be denoted as $\mathcal{O}(\sum_{a=0}^{A-1} u_a^{actor} u_{a+1}^{actor} + \sum_{c=0}^{C-1} u_c^{critic} u_{c+1}^{critic})$. Thus, for K time slots and $M+N$ agents, the computational complexity of the inner A-MADDPG algorithm is $\mathcal{O}(K(M+N)\mathcal{Y})$, in which $\mathcal{Y} = \sum_{a=0}^{A-1} u_a^{actor} u_{a+1}^{actor} + \sum_{c=0}^{C-1} u_c^{critic} u_{c+1}^{critic}$. To sum up, the overall computational complexity of the proposed algorithm is $\mathcal{O}(L_{ep}L_{frame}K(M+N)\mathcal{X}\mathcal{Y})$.

Note that the DDQN algorithm and the centralized training process of the MADDPG algorithm are implemented on the UGCS. In the MADDPG algorithm, the centralized training process is uploaded to the UGCS to ease the burden of the UAV and user agents, and then in the distributed execution process, the well-trained weights are downloaded from the UGCS to enable each agent to take the action based on its own observation [24], [46]. Therefore, the computational burden borne by the UAV is within its acceptable range due to the powerful processing capability of the ground control center.

V. Simulation Results

In this section, simulations results are provided to validate the effectiveness of the proposed algorithm. The simulated region of the wireless caching network is a square area of 2 km $\times$ 2 km. The terrestrial BS is located at the center of the square. The simulated number of users is 15 and for the sake of simplicity, these users are equally divided into 3 clusters according to the type of requested services, and users in the same cluster have similar preference for cache files. As in the previous work [47], the length of the time slot τ is set as $\tau = 1$ s. The adaptive moment estimation (ADAM) optimizers [48] are adopted with learning rate of 0.0001. For all the neural networks, the ReLU activation function is utilized for hidden layers and the linear activation function is utilized for output layers. Other parameters including hyper parameters of the ATMADDPG and DDQN frameworks used throughout the simulations are summarized in Table II, unless otherwise specified.

We first confirm the convergence performance of our proposed 2TDRL algorithm in Fig. 4. In this figure, we plot the system throughput achieved by our proposed scheme versus the number of episodes. For comparison purpose, the performances achieved by three benchmark schemes are also plotted, namely, 1) the 2TDRL without cache placement (2TDRLWC) scheme, in which only the user association and UAV trajectory planning are optimized without considering the cache placement. The cached files before flight are randomly chosen; 2) the MADDPG scheme, in which the cache placement in the large timescale is optimized by the same policy as adopted in 2TDRL, while the user association and trajectory planning in the small time scale are updated by the MADDPG algorithm [49]; 3) the distributed DQN (d-DQN) scheme, in which the cache placement is optimized by the same policy as in 2TDRL, while the user association and trajectory planning are updated by distributed DQN algorithm [23], where each agent can collaborate with its neighboring agents to learn the system configuration policy; 4) the multi-agent twin delayed deep deterministic (MATD3) policy gradient scheme, in which the cache placement is optimized by the same policy as in 2TDRL, while the user association and trajectory planning are updated by the MATD3 algorithm [50].

We first see that, as the number of episodes increases, the system throughput achieved by our proposed solution approaches the certain value, verifying the convergence of our proposed

TABLE II
Simulation Parameters

System bandwidth	1 MHz
Transmit power of UAVs	10 dBm
Transmit power of BS	20 dBm
Noise power spectral density	-174 dBm/Hz
Pathloss exponent	2.7
Flight cycle	100 s
Maximum UAV speed	50 m/s
Minimum UAV speed	10 m/s
Flight altitude	100 m
Average velocity of users	4 m/s
Total number of content files	30
Cache storage capacity of UAVs	20
Number of hidden layers	2
Number of neurons per layer	$\{256, 128\}$
Discount factor γ	0.99
Experience replay buffer size $\mathcal{E}$	50000
Experience replay buffer size $\mathcal{G}$	10000
Mini-batch size N_B	32
Soft update rate	0.005
safe distance d_{col}	10 m
Penalty of violating constraints	50

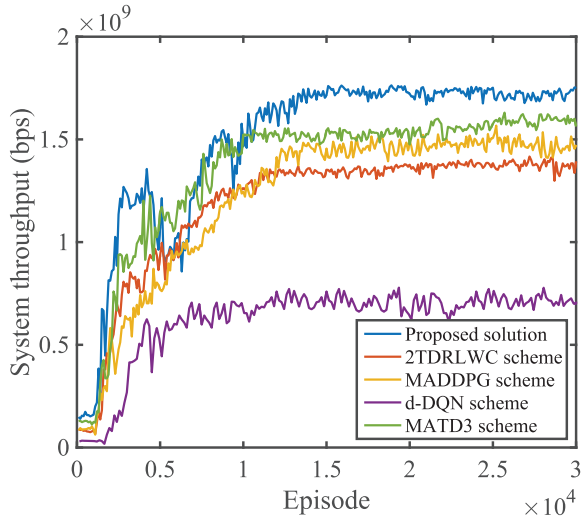


Fig. 4. Sum Reward achieved by all agents per episode in the training stage. Each data point is the maximum value in every 100 episodes.

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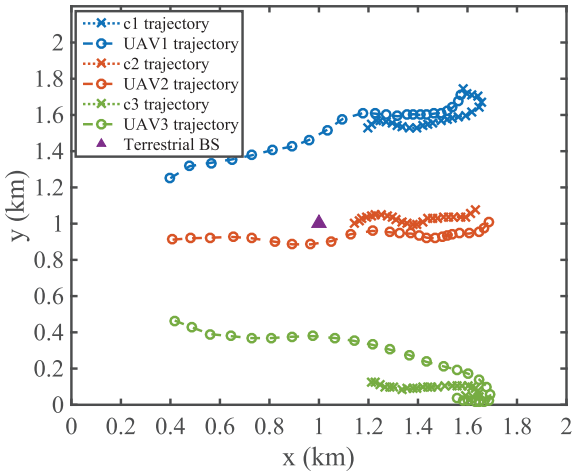


Fig. 5. Trajectories of UAVs designed by our proposed scheme.

scheme. Moreover, our proposed scheme achieves the best performance. By comparing our scheme with the 2TDRLWC scheme, the effectiveness of reasonable cache placement strategy in improving the system throughput can be verified, in which the throughput is increased by 16.8%. In each iteration, UAVs learn cache placement policies based on the file popularity (requested times of each file) in previous K time slots, and thus, can better meet the needs of users. We can also see that, our proposed scheme outperforms the MATD3 scheme, and the MATD3 scheme outperforms the MADDPG scheme. The reason for the first observation is that the attention mechanism adopted in our scheme enables the agents to pay more attention to the valuable agents and obtain more useful information. Therefore, the learning efficiency can be improved and the agents can converge to the optimal policies faster. The reason for the second observation is that the double Q networks incorporated in the MATD3 scheme enables the smaller Q-value to be chosen as the target value, which can efficiently relieve the overestimation error in MADDPG. Therefore, the learning efficiency can be improved and the agents can converge to the optimal policies faster. In addition, the throughput of d-DQN scheme is shown to be the minimum due to its limited coordinated training and quantization of action.

In Fig. 5, we plot the trajectories of UAVs designed by our proposed scheme. The initial locations of the UAVs in the simulation section is predetermined and fixed, while the final locations of the UAVs are the locations at the end of the flight cycle. As referred in [51], [52], the energy consumption during the whole flight cycle T is within the UAVs' acceptable battery capacity. The trajectories of three clustered users are also provided for reference. We can see that, although the initial locations of UAVs are far away from the clustered users, our proposed scheme can enable the UAVs to choose the corresponding user clusters automatically, and then move along with the associated user clusters to provide caching services. This implies that our proposed scheme can leverage the UAV mobility to further improve the performance of the cache-enabled UAV networks.

Fig. 6(a) plots the system throughput versus the number of UAVs N for different schemes. It can be observed that the system throughputs achieved by all the schemes increase with the number of UAVs. This is caused by two reasons: the increase of the number of UAVs 1) leads to the increase in the number of files that can be cached; 2) enables the UAVs to move along with the users, thus providing the users with timely and higher-quality services than the terrestrial BS. Moreover, the growth trend goes to be flat, which is due to the limited improvement of system throughput when the number of users remains unchanged. We can also see that the performance gap between our proposed scheme and the 2TDRLWC scheme is small when N is small, while growing larger when N becomes relatively large. The is because that when N is small, the number of files cached in UAVs are rather limited, resulting in the smaller performance gain brought by the cache placement strategy compared to the attention mechanism. However, as the number of UAVs increases, the impact of cache placement strategy on the overall throughput performance gradually becomes significant.

Fig. 6(b) examines the effect of the number of UAVs N on the throughput ratio of UAVs for different schemes. The throughput ratio of UAVs ρ_{UAV} is defined as the throughput achieved by UAVs divided by the total system throughput, namely $\rho_{UAV} = \sum_k R_{tot}(k) \sum_k R_{uav}(k)$. It is obvious that the throughput ratio of UAVs increases with N. Moreover, we can observe that ρ_{UAV} in the MADDPG and MATD3 schemes are larger than that in the 2TDRLWC scheme. This is because that the optimized cache placement strategy in these schemes enables UAVs to better satisfy the user requirements, leading to more users accessing to the UAVs for cache services. However, the cache placement strategy adopted in the 2TDRLWC scheme is random, leading to more users accessing to the terrestrial BS for cache services.

Except for investigating the influence of N on the throughput ratio of UAVs, we further examine the impact of N on the average throughput per UAV in Fig. 6(c). By comparing the results in Figs. 6(b) and 6(c), we can see that, although the increase in N can bring benefits to the throughput ratio of the UAVs, the average throughput per UAV decreases. This is because, as N increases, the number of users served by each UAV decreases, resulting in the decrease of the average throughput per UAV.

Fig. 7 depicts the impact of UAVs' transmission power on the system throughput for $N = 1$, $N = 3$, and $N = 5$. It can be observed that the system throughput increases with UAV's transmission power. The reason is that based on (11) and (15), the increase of UAVs' transmission power contributes to attracting more users and achieving higher transmission rate, thus, enhancing the system throughput.

In Fig. 8(a), we examine the effect of the cluster number on the system throughput for different schemes. We consider that the number of users in each cluster is fixed as 5. It can be observed that the system throughput increases as the number of clusters increases. Moreover, the performance gap between our proposed scheme and the 2TDRLWC scheme also increases with the number of clusters. The reason is that more user clusters call for more kinds of files, which highlights the importance of the reasonable cache placement strategy.

Aside from the dependence on the number of clusters, we further examine the effect of the number of users in each cluster

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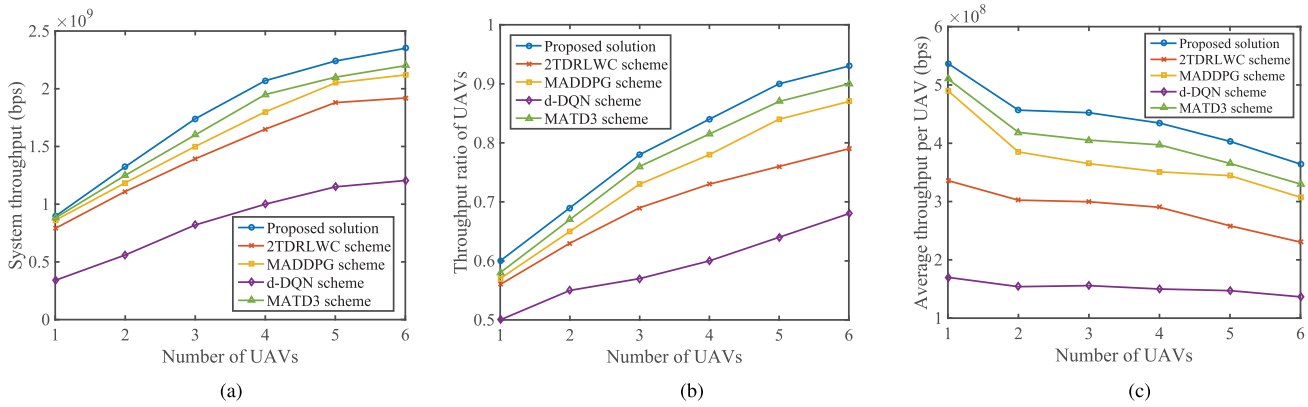


Fig. 6. Effect of the number of UAVs N on the system throughput, the throughput ratio of UAVs and the average throughput per UAV for different schemes. (a) System throughput versus N (b) Throughput ratio of UAVs versus N. (c) Average throughput per UAV versus N.

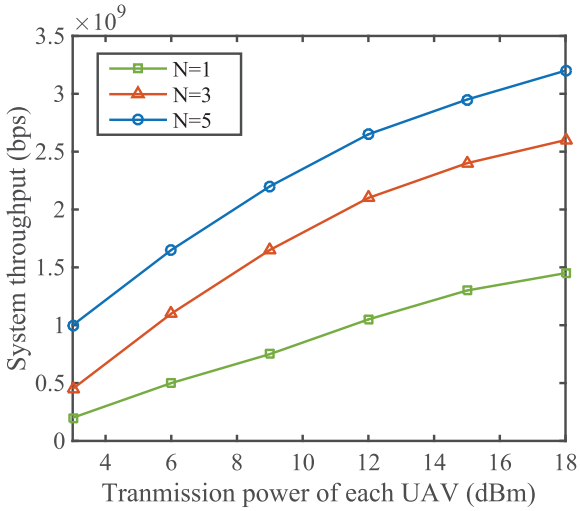


Fig. 7. Effect of UAVs' transmission power on the system throughput.

on the throughput performance in Fig. 8(b). In this figure, we consider that the number of clusters is fixed as 3. We note that, compared with the number of clusters, the number of users in each cluster has more significant impact on the system throughput. This is because in our scheme, the trajectories of UAVs are determined by the center of each cluster. Note that users in the same cluster have similar preference for requested files, so there is no obvious variations on the kinds of requested files when the number of clusters keeps unchanged. Therefore, when the number of users in each cluster increases, UAVs can provide nearby service for more users, leading to the increase in the system throughput. However, when the number of clusters increases, the trajectories of UAVs are impacted by multiple clusters, which may degrade the service quality that UAVs provide.

Furthermore, in order to verify the performance advantages of our air-ground cooperative architecture that combines both the UAVs and the terrestrial BS, we introduce the terrestrial BS-only scenario as a comparison and investigate the effect of user number on the system throughput in Fig. 8(c). It is obvious that the performance of the UAVs-BS architecture significantly outweighs the performance of the BS-only architecture. The reason is that compared with the BS-only scenario, the utilization of both the UAVs and the terrestrial BS can provide users with prompt, all-round and seamless services, in which the UAVs fly close to the users through trajectory planning to provide convenient and high-quality services, and the terrestrial BS serves as a supplement to make up for the lack of UAVs' uncached files.

In Fig. 9, we examine the influence of the cache storage capacity of each UAV on the throughput of the UAVs for different values of content popularity factor δ. It can be observed that the throughput of the UAVs increases with the cache storage capacity of each UAV, due to that the cache hit probability increases as the cache storage capacity of each UAV increases. Moreover, it is shown that the throughput of UAVs can be improved when the popularity factor δ increases and the cache storage capacity of each UAV is in the medium region. The improvement diminishes when the cache storage capacity of each UAV becomes relatively large. This is because when the cache storage capacity of each UAV increases, all the files in the library $\mathcal{F}$ can be cached at the UAVs. In such case, the cache hit probability is maximized and the distribution of users' file requests cannot impact the throughput of UAVs.

Finally, we examine the impact of the cache storage capacity of each UAV on the caching efficiency in Fig. 10. The caching efficiency η_c is defined as the throughput of UAVs brought by caching one single file. It can be observed that, as the cache storage capacity of each UAV increases, the caching efficiency firstly increases and then decreases, showing that enhancing the cache storage capacity of each UAV may not always be the optimal option for the system operators, especially when the storage module can be prohibitively expensive. Such an observation can be utilized as a guidance to set the storage capacity as the value of achieving maximum η_c.

Proposition 1: The optimal value of the cache storage capacity for achieving the maximization of the caching efficiency is near the value of the user number M.

Proof: UAVs' caching efficiency η_c is given by

$$\eta_c = \frac{\sum_{k=1}^{K}\sum_{m=1}^{M} R_m^{uav}(k)}{C} = \frac{\sum_{k=1}^{K}\sum_{n=1}^{N}\sum_{m=1}^{M}\sum_{f=1}^{F}\alpha_{n,m,f}^{k}\beta_{n,f}B_m^k \log_2\left(1+\frac{p_{n,m}^k\left|H_{n,m}^k\right|^2}{\sigma_{n,m}^k}\right)}{C}, \tag{41}$$

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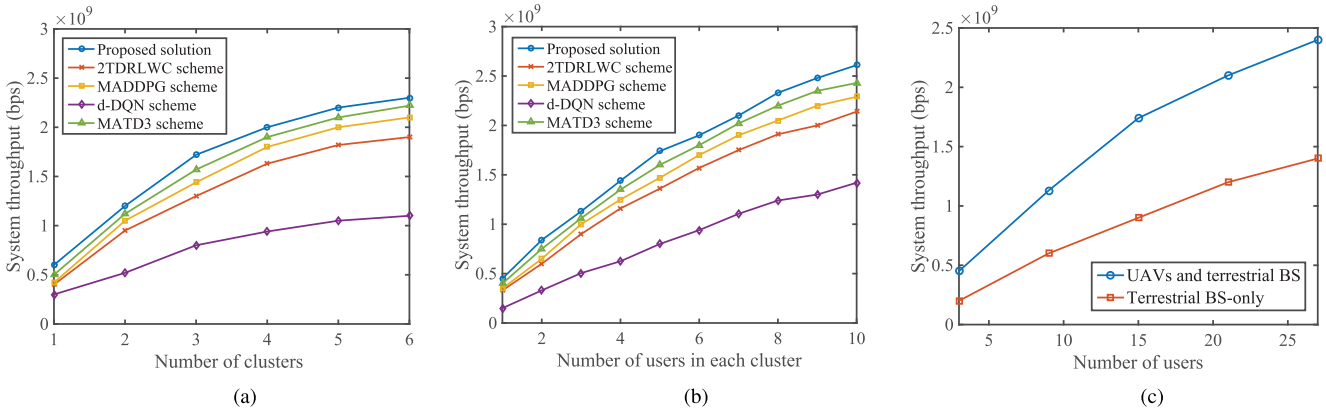


Fig. 8. Effect of user number on the system throughput. (a) System throughput versus the number of user clusters. (b) System throughput versus the number of users in each cluster. (c) Throughput comparison between the UAVs-BS architecture and the BS-only architecture.

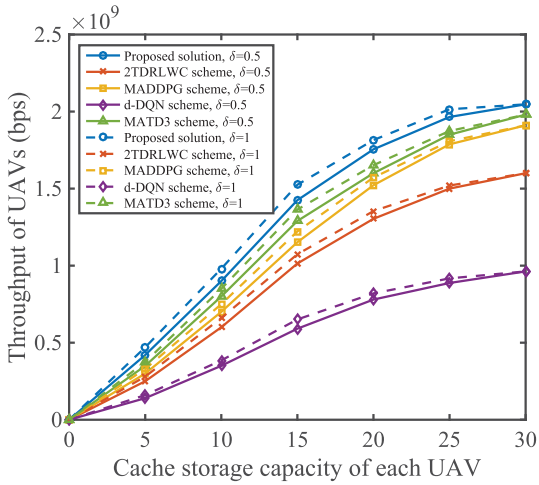


Fig. 9. Throughput of UAVs versus the cache storage capacity of each UAV.

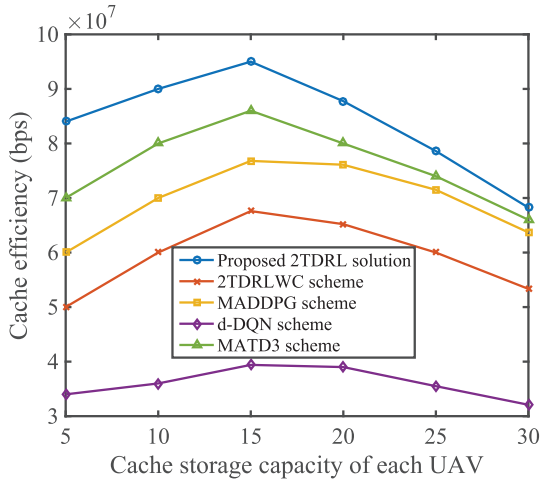


Fig. 10. Caching efficiency versus the cache storage capacity of each UAV.

where C is the average cache storage capacity of the UAVs. In order to facilitate the analysis, we consider two ideal assumptions: 1) the files cached in the UAVs have high probability to be requested by users, and 2) users will access to the UAVs as long as their requested files can be found in the UAVs. Note that the first assumption can be achieved by the well-designed cache placement strategy.

- When the condition $0 < C \leq M$ satisfies, as C increases, the number of users accessing to the UAVs increases, and $\sum_{n=1}^{N}\sum_{m=1}^{M}\sum_{f=1}^{F}\alpha_{n,m,f}^{k}\beta_{n,f}$ also increases proportionally. Moreover, the UAVs can follow users closely and provide higher-quality services, thus achieving higher throughput compared to that of the terrestrial BS. Therefore, η_c increases with C.
- When the condition $M < C \leq F$ satisfies, as C increases, the additional increasing number of users accessing to the UAVs becomes smaller, and the growth rate of $\sum_{n=1}^{N}\sum_{m=1}^{M}\sum_{f=1}^{F}\alpha_{n,m,f}^{k}\beta_{n,f}$ also becomes smaller. Therefore, η_c starts to slowly decrease with C.
- When the condition $C > F$ satisfies, the whole library $\mathcal{F}$ can be cached in the UAVs, and the increase of C will not improve the UAVs throughput anymore. Therefore, η_c decreases with C.

To sum up, under the approximately ideal assumptions, the optimal value of the cache storage capacity for achieving the maximization of the caching efficiency is near the value of the user number M. The proof has been completed.

VI. Conclusion

In this article, we considered a UAV-assisted edge caching network, where UAVs serve as aerial BSs, together with the terrestrial BS to provide caching service for users. We propose a two-timescale-DRL-based user association, cache placement and trajectory planning scheme to maximize the system throughput. The optimization problem is decomposed into a cache placement subproblem with long updating cycle and a user association and UAV trajectory planning subproblem with short updating cycle. To tackle the former subproblem, the DDQN method is applied to autonomously update the cache placement based on

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the previously acquired experience. To tackle the latter subproblem, we propose an attention-based MADDPG algorithm, where users and UAVs work as two different kinds of agents to learn the cooperation among each other for maximizing the long-term accumulative rewards. A 2TDRL algorithm is proposed to solve these two subproblems iteratively. Extensive numerical results are provided to verify the performance of our proposed solution compared with the state-of-art schemes, showing how our scheme can be useful for practical UAV-assisted edge caching networks. In the future work, we would like to design novel cache-enabled UAV networks supporting massive number of users and the users' mobility. As such, the impacts of key parameters on the cache-enabled UAV networks, such as the bandwidth allocation among users, the altitude adjustment of UAVs, as well as the sensing-assisted obstacle avoidance should be taken into account to further enhance the system performance.

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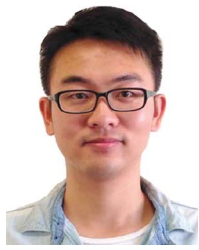
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